

MuLE · MULTILINGUAL LANGUAGE EQUILIBRIUM

DPO Pipeline Structure

How self-generated answers in English, French and Portuguese become preference pairs that teach the M-Thinker Cold-Start SFT 1.5B model to reason in the input language.

STAGE 1 Answer generation [RL Train Answers Generation/](#)

vLLM samples $N = 8$ answers per question per language (en · fr · pt) from XueZhang-bjtu/1.5B-cold-start-SFT. Each stored candidate carries a prepended `<think>\n` tag.

iter1_questions.jsonl — triples (en, fr, pt) per question



STAGE 2 Preference-pair construction [dataset_filter.py](#) [rewards.py](#)

1 · SCORE EVERY ANSWER

acc $\in \{0, 1\}$ `acc_compute_score` — last `\boxed{}` vs ground truth (rewards.py)**lc** $\in \{1, 0, -1\}$ `fastText` lid.176, 350-char windows; a confident top-1 label outside {en, fr, pt, es} (e.g. zh leakage) counts as off-language, otherwise confidence is renormalised over the candidates; think and answer sections judged separately; English function-word check on non-English text**format** $\in \{0, 1\}$ exactly one `</think>`, length > 1000, `\boxed{}` after the close**bt count**

backtracking-signal regex hits (currently unused in ranking)

2 · BUCKET ANSWERS

Chosen-eligible $\text{acc} = 1 \cdot \text{lc} = 1 \cdot \text{format} = 1$ **Rejected** $\text{acc} = 0$ (any lc), or $\text{acc} = 1$ with $\text{lc} = 0$ (right answer, wrong language)**Excluded — never trained on**correct but $\text{lc} = -1$ (ambiguous) or broken format

3 · PICK PIVOT LANGUAGE

Per question, the language with the highest **chosen-eligible** rate ($\text{acc} \wedge \text{lc} \wedge \text{format}$ — same test as the chosen pool) becomes the **pivot**; ties go to English. The same rate feeds `total_consensus` and the difficulty tiers below, so a pivot with correct rate > 0 always has usable reference answers. Questions with zero chosen-eligible answers anywhere are dropped (`skipped_no_correct_answers`).

4 · RANK CHOSEN BY CROSS-LINGUAL AGREEMENT

get_latex()

LaTeX from the final reasoning segment (after the last backtrack signal)

**chrF (n = 6)**

character n-gram F-score vs each correct pivot answer (pivot answers compare against the other two languages)

**max alignment**